

Notes: Pohl, Schmedders, and Wilms (JFE, 2021)

Asset Pricing with Heterogeneous Agents and Long-Run Risk

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1 Big picture

- **Question.** What happens to long-run risk asset pricing once we abandon the representative-agent assumption and allow investors to disagree about the persistence of long-run consumption risk?
- **Setting.** The paper studies an infinite-horizon complete-markets endowment economy with recursive preferences. The aggregate endowment follows the standard Bansal–Yaron long-run risk process, but investors differ in their beliefs about the persistence parameter ρ_x .
- **Core challenge.** In heterogeneous-agent economies with recursive utility, equilibrium allocations are no longer functions of the exogenous state alone. The wealth distribution becomes an endogenous state variable, so standard representative-agent solution methods break down.
- **Main theoretical contribution.** The paper reformulates the social planner’s first-order conditions in a recursive way by introducing time-varying **Negishi weights**. These weights summarize the relative trend in agents’ consumption shares and turn the equilibrium problem into a tractable recursive system.
- **Main economic contribution.** Belief differences generate two distinct motives for trade:
 - a **speculation motive**: agents trade because they assign different probabilities to future states;
 - a **risk-sharing motive**: with Epstein–Zin preferences, agents hedge states that are especially bad for their continuation utility.
- **Why this matters.** Under CRRA preferences, the speculation motive alone implies the classical market-selection result: in the long run, the agent with correct beliefs dominates. Under Epstein–Zin preferences, the risk-sharing motive can offset that force, so even agents with incorrect beliefs can keep a meaningful wealth share for a long time and materially affect prices.
- **Baseline calibration.** The quantitative exercise stays as close as possible to Bansal and Yaron’s Case I long-run risk model, *without* stochastic volatility. The only change is heterogeneity in beliefs:

$$\rho_x^A = 0.99, \quad \rho_x^B = 0.96,$$

with agent A holding the correct beliefs.

- **Main quantitative result.** Small belief differences create large endogenous movements in consumption and wealth shares. These shifts add a new priced risk to the model and substantially improve its asset-pricing performance.
- **Key moments.**
 - The heterogeneous-agent model generates an annualized equity premium of about **5.42%**.
 - The standard deviation of the expected risk premium is about **5.73%**, close to the empirical benchmark in Martin (2017).
 - The standard deviation of the log price-dividend ratio rises to about **0.38**, much closer to the data than the representative-agent benchmark.

- **Predictability result.** The model increases return predictability and reduces the excessive predictability of consumption and dividend growth relative to the representative-agent long-run risk model. It also generates **stronger return predictability in recessions**, consistent with the data.
- **Model comparison result.** A representative-agent model with exogenous time-varying persistence does not reproduce the same strong, state-dependent variation in risk premia. The reason is that it misses the endogenous redistribution channel operating through wealth shares.
- **Economic takeaway.** Disagreement about a parameter that is difficult to estimate—the persistence of long-run growth risk—is enough to endogenously generate excess volatility, counter-cyclical risk premia, and recession-dependent predictability in a parsimonious extension of the long-run risk model.

2 Model environment and key objects

2.1 The general heterogeneous-agent economy

Time is discrete, $t = 0, 1, 2, \dots$, and the exogenous state is y_t . Agent $h \in \mathcal{H} = \{1, \dots, H\}$ chooses individual consumption $C^h(y^t)$, where $y^t = (y_0, \dots, y_t)$. Aggregate consumption is $C(y^t)$, and market clearing is

$$\sum_{h=1}^H C^h(y^t) = C(y^t). \quad (1)$$

Interpretation. This is a pure endowment economy: asset prices are determined by how agents split the aggregate stream across states and time.

2.2 Recursive utility and heterogeneous beliefs

Each agent has recursive utility

$$U^h(\{C^h\}_t) = F^h\left(C^h(y^t), R_t^h\left[U^h(\{C^h\}_{t+1})\right]\right), \quad (2)$$

with certainty equivalent

$$R_t^h[x] = (G^h)^{-1}\left(E_t^h[G^h(x)]\right). \quad (3)$$

Agents can disagree about the conditional law of next period's state. If $P_{t,t+1}$ is the true conditional distribution and $P_{t,t+1}^h$ is agent h 's subjective conditional distribution, then

$$E_t^h[x] = E_t\left[x \frac{dP_{t,t+1}^h}{dP_{t,t+1}}\right].$$

Interpretation. The Radon–Nikodym derivative is the formal device that converts true probabilities into agent-specific subjective probabilities. It is the source of disagreement in the model.

2.3 Complete markets and the social planner

Because markets are complete, equilibrium can be characterized through a social planner who chooses consumption processes $\{C^h\}_0$ to maximize

$$SP(\{C\}_0; \lambda) = \sum_{h=1}^H \bar{\lambda}^h U^h(\{C^h\}_0), \quad (4)$$

subject to the resource constraint (1). The vector $\lambda = (\bar{\lambda}^1, \dots, \bar{\lambda}^H)$ contains positive Negishi weights.

Interpretation. In the representative-agent model, equilibrium allocations depend only on the exogenous state. Here they depend on exogenous states *and* on the endogenous wealth distribution summarized by the time-varying planner weights.

2.4 The long-run risk endowment process

For the quantitative application, aggregate consumption and dividends follow the standard long-run risk system:

$$\Delta c_{t+1} = \mu_c + x_t + \eta_{c,t+1}, \quad x_{t+1} = \rho_x x_t + \eta_{x,t+1}, \quad \Delta d_{t+1} = \mu_d + \Phi x_t + \eta_{d,t+1}. \quad (16)$$

The shocks $\eta_{c,t+1}$, $\eta_{x,t+1}$, and $\eta_{d,t+1}$ are independent normals with standard deviations σ , $\phi_x \sigma$, and $\phi_d \sigma$, respectively.

Interpretation. The key state x_t is a highly persistent component in expected growth. Long-run risk only matters for asset pricing if ρ_x is very close to one, which is precisely why disagreement about ρ_x is economically plausible.

2.5 Belief heterogeneity about persistence

Agents agree on all parameters except the persistence of the long-run risk state. Agent h 's subjective law for x_{t+1} is

$$dP_{t,t+1}^h = \frac{1}{\sqrt{2\pi}\sigma_x} \exp \left[-\frac{1}{2} \left(\frac{x_{t+1} - \rho_x^h x_t}{\sigma_x} \right)^2 \right]. \quad (17)$$

The paper's baseline calibration sets

$$\psi^A = \psi^B = 1.5, \quad \gamma^A = \gamma^B = 10, \quad \delta^A = \delta^B = 0.998,$$

$$\mu_c = \mu_d = 0.0015, \quad \sigma = 0.0078, \quad \Phi = 3, \quad \phi_d = 4.5, \quad \phi_x = 0.044,$$

and varies only the belief about persistence:

$$\rho_x^A = 0.99, \quad \rho_x^B = 0.96.$$

Interpretation. The paper is deliberately parsimonious. It changes only one parameter relative to the standard calibration, so any improvement in asset-pricing performance is attributable to belief

heterogeneity rather than to extra exogenous shocks.

2.6 Key economic objects

The paper tracks four state variables and outcomes:

- the long-run growth state x_t ;
- the endogenous consumption share $s_t^h = C_t^h/C_t$;
- the endogenous wealth share W_t^h/W_t ;
- asset prices such as the market price-dividend ratio and expected risk premia.

Interpretation. The essential new state variable relative to the representative-agent model is the **wealth distribution**. All of the paper's main asset-pricing implications come from how shocks move this distribution.

3 Models and analytical specifications (all equations + interpretation)

3.1 First-order conditions and recursive weights

The planner's optimality conditions imply, for each $t \geq 0$,

$$\lambda_t^h F_1^h(C_t^h, R_t^h[U_{t+1}^h]) = \lambda_t^1 F_1^1(C_t^1, R_t^1[U_{t+1}^1]), \quad (5)$$

with initial condition

$$\lambda_0^h = \bar{\lambda}^h, \quad (6)$$

and recursive law of motion

$$\frac{\lambda_{t+1}^h}{\lambda_{t+1}^1} = \frac{\Lambda_{t+1}^h}{\Lambda_{t+1}^1} \frac{\lambda_t^h}{\lambda_t^1}, \quad h \in \{2, \dots, H\}, \quad (7)$$

where

$$\Lambda_{t+1}^h = F_2^h(C_t^h, R_t^h[U_{t+1}^h]) \cdot \frac{(G^h)'(U_{t+1}^h)}{(G^h)'(R_t^h[U_{t+1}^h])} \cdot \frac{dP_{t,t+1}^h}{dP_{t,t+1}}. \quad (8)$$

Interpretation. The weights λ_t^h are not fixed Pareto weights. They evolve with beliefs, continuation values, and marginal utilities. They are the endogenous state variables that restore recursiveness.

3.2 Epstein–Zin specialization

For Epstein–Zin preferences,

$$F^h(c, x) = \left[(1 - \delta^h) c^{\rho^h} + \delta^h x^{\rho^h} \right]^{1/\rho^h}, \quad (9)$$

$$G^h(x) = x^{\alpha^h}, \quad (10)$$

where

$$\rho^h = 1 - \frac{1}{\psi^h}, \quad \alpha^h = 1 - \gamma^h.$$

Let $s_t^h = C_t^h/C_t$ denote the consumption share and $v_t^h = V_t^h/C_t$ the normalized value function. Market clearing implies

$$\sum_{h=1}^H s_t^h = 1. \quad (11)$$

The normalized value function satisfies

$$v_t^h = \left[(1 - \delta^h)(s_t^h)^{\rho^h} + \delta^h R_t^h \left[(v_{t+1}^h e^{\Delta c_{t+1}})^{\rho^h} \right] \right]^{1/\rho^h}. \quad (12)$$

Define the normalized weight

$$\tilde{\lambda}_t^h = \frac{\lambda_t^h}{(v_t^h)^{\rho^h - 1}}.$$

Then the consumption share is pinned down by

$$\tilde{\lambda}_t^h (1 - \delta^h)(s_t^h)^{\rho^h - 1} = \tilde{\lambda}_t^1 (1 - \delta^1)(s_t^1)^{\rho^1 - 1}. \quad (13)$$

The normalized weights evolve according to

$$\frac{\tilde{\lambda}_{t+1}^h}{\tilde{\lambda}_{t+1}^1} = \frac{\tilde{\Lambda}_{t+1}^h}{\tilde{\Lambda}_{t+1}^1} \frac{\tilde{\lambda}_t^h}{\tilde{\lambda}_t^1}, \quad (14)$$

with

$$\tilde{\Lambda}_{t+1}^h = \delta^h e^{\rho^h \Delta c_{t+1}} \frac{dP_{t,t+1}^h}{dP_{t,t+1}} \left(\frac{(v_{t+1}^h e^{\Delta c_{t+1}})^{\alpha^h - \rho^h}}{R_t^h [(v_{t+1}^h e^{\Delta c_{t+1}})^{\alpha^h - \rho^h}]} \right). \quad (15)$$

Interpretation. Relative to discounted expected utility, the new state variables now depend on continuation values. This is exactly why recursive utility makes wealth-distribution dynamics quantitatively important.

3.3 Consumption-share dynamics under heterogeneous beliefs

When agents share the same EIS ψ , the two-agent consumption-share ratio can be written as

$$\frac{s_{t+1}^A}{s_{t+1}^B} = \frac{s_t^A}{s_t^B} \left(\frac{dP_{t,t+1}^A}{dP_{t,t+1}^B} \right)^\psi \left(\frac{v_{t+1}^A / R_t^A(v_{t+1}^A)}{v_{t+1}^B / R_t^B(v_{t+1}^B)} \right)^{1 - \gamma\psi}. \quad (18)$$

The first term is the **CRRA term**; the second is the extra **Epstein–Zin term**.

Interpretation. This decomposition is the paper's core mechanism.

- The first term is pure **speculation**: agents trade because they disagree about next period's state probabilities.

- The second term is **risk sharing**: agents trade because they care about how states affect continuation utility, not just one-period consumption.

3.4 Discrete-state mean–variance intuition

To explain the risk-sharing channel, the paper analyzes a stylized two-state Markov version of the model. Define

$$\bar{v}_t^h = \frac{(v_t^h)^{\rho^h}}{\rho^h}.$$

Then

$$\bar{v}_t^h = \frac{(1 - \delta^h)(s_t^h)^{\rho^h}}{\rho^h} + \delta^h E_t^h \left[(\bar{v}_{t+1}^h)^{\alpha^h / \rho^h} e^{\alpha^h \Delta c_{t+1}} \right]^{\rho^h / \alpha^h}. \quad (22)$$

If $\bar{v}_{t+1}^h$ is approximately lognormal, the continuation value can be approximated by

$$\bar{v}_t^h \approx \frac{(1 - \delta^h)(s_t^h)^{\rho^h}}{\rho^h} + \delta^h k_t^h E_t^h(\bar{v}_{t+1}^h) - \delta^h k_t^h \frac{\rho^h - \alpha^h}{2\rho^h} \frac{\text{Var}_t^h(\bar{v}_{t+1}^h)}{E_t^h(\bar{v}_{t+1}^h)}, \quad (23)$$

where $k_t^h = E_t^h(e^{\alpha^h \Delta c_{t+1}})^{\rho^h / \alpha^h}$.

Interpretation.

- With CRRA preferences, $\rho^h = \alpha^h$, so the variance term disappears.
- With Epstein–Zin preferences and $\gamma > 1/\psi$, continuation-value variance directly lowers welfare.
- Therefore an agent may rationally give up expected future wealth in exchange for insurance against continuation-value risk.

This is why agents with incorrect beliefs do not necessarily disappear quickly in the Epstein–Zin economy.

3.5 Wealth shares and asset pricing

Individual wealth is the claim to an agent’s future consumption stream. For Epstein–Zin utility,

$$\frac{W_t^h}{C_t^h} = \frac{1}{1 - \delta^h} \left(\frac{V_t^h}{C_t^h} \right)^{\rho^h}. \quad (25)$$

The stochastic discount factor for agent h is

$$M_{t+1}^h = \delta^h e^{(\rho^h - 1)\Delta c_{t+1}} \left(\frac{s_{t+1}^h}{s_t^h} \right)^{\rho^h - 1} \frac{(v_{t+1}^h e^{\Delta c_{t+1}})^{\alpha^h - \rho^h}}{R_t^h [(v_{t+1}^h e^{\Delta c_{t+1}})^{\alpha^h - \rho^h}]}. \quad (26)$$

Interpretation. Asset prices now inherit risk from two sources:

- the usual long-run risk in aggregate consumption growth;
- the endogenous redistribution risk coming from time variation in s_t^h and W_t^h/W_t .

The paper’s main quantitative message is that this second source is both large and strongly countercyclical.

3.6 Why disagreement about ρ_x is plausible

The paper gives two empirical reasons.

First, if an investor tries to estimate ρ_x from finite historical data, the estimates are noisy. The paper studies:

$$x_{t+1} = \mu_x + \rho_x x_t + \sigma_x \eta_{x,t+1}, \quad (19)$$

and, when x_t is unobserved, the state-space system

$$\Delta c_{t+1} = \mu_c + x_t + \sigma \eta_{c,t+1}, \quad x_{t+1} = \rho_x x_t + \sigma_x \eta_{x,t+1}. \quad (20)$$

Second, using professional forecasts, the paper estimates

$$\Delta c_{t,i}^i = A_i + \beta \Delta c_{t,i} + \beta_i \Delta c_{t,i} + K_t + e_{t,i}, \quad (21)$$

where $\beta + \beta_i$ measures each forecaster’s persistence estimate.

Interpretation. The parameter on which the entire long-run risk model hinges is also one of the hardest parameters to estimate precisely. So disagreement is not an arbitrary modeling device; it is an economically natural feature.

3.7 Representative-agent alternative with time-varying persistence

To show what is special about heterogeneity, the paper compares its model with a representative-agent economy in which persistence itself follows an exogenous process:

$$\Delta c_{t+1} = \mu_c + x_t + \sigma \eta_{c,t+1}, \quad x_{t+1} = \rho_{x,t} x_t + \sigma_x \eta_{x,t+1}, \quad (28a-b)$$

$$\rho_{x,t+1} = \mu_\rho(1 - \nu) + \nu \rho_{x,t} + \sigma_{\rho,x} \eta_{x,t+1} + \sigma_\rho \eta_{\rho,t+1}, \quad (28c)$$

$$\Delta d_{t+1} = \mu_d + \Phi x_t + \phi_d \sigma \eta_{d,t+1}. \quad (28d)$$

The “aggregate persistence” in the heterogeneous-agent economy is defined as

$$\rho_{x,t}^{\text{Agg}} = s_t^B \rho_x^B + (1 - s_t^B) \rho_x^A. \quad (29)$$

Interpretation. This alternative model matches time variation in persistence mechanically, but it misses the fact that the mapping from today’s state x_t to tomorrow’s effective persistence is endogenous and depends on how wealth gets redistributed across agents.

4 Mechanism, calibration, and what makes the quantitative argument credible

4.1 Why the calibration is persuasive

The calibration is intentionally conservative:

- it starts from Bansal–Yaron Case I, not a newly designed model;
- it removes stochastic volatility, so any time variation in risk premia is not imported by construction;
- it changes only one economically meaningful parameter: beliefs about ρ_x .

Interpretation. This makes the paper’s quantitative improvements hard to dismiss as a product of overfitting. The model gets more with less.

4.2 Speculation motive versus risk-sharing motive

The paper’s main economic mechanism is the interaction between two forces:

- **Speculation motive.** If investors disagree about which state is more likely tomorrow, each wants to hold more claims to the states she thinks are underpriced.
- **Risk-sharing motive.** If investors have Epstein–Zin preferences and care about long-run utility risk, then an investor who especially dislikes low-growth states is willing to pay another investor to insure her against those states.

Interpretation. The first force looks like standard disagreement trading. The second is what changes everything. It makes wealth dynamics depend not only on who is right, but also on who is more willing to warehouse long-run risk.

4.3 Why incorrect-belief investors can survive

Under CRRA preferences, wrong-belief investors lose in the long run because only the speculation motive is present. Under Epstein–Zin preferences, however, an investor who believes in lower persistence may be willing to bear recession risk in exchange for compensation in good times. That can produce a persistent transfer of wealth toward that investor even if she is wrong about the true law.

Interpretation. This result is crucial for asset pricing. If wrong-belief investors disappeared quickly, heterogeneity would not matter much for long-run moments. The paper shows the opposite: disagreement can have long-lasting equilibrium price effects.

4.4 Why the model generates countercyclical risk premia

When the economy receives negative long-run risk shocks, the wealth share of the investor who believes in *higher* persistence rises because the insurance against low-growth states pays off. That investor also demands a higher premium for holding risky cash flows. As a result:

- expected risk premia rise in recessions;
- price-dividend ratios become more volatile;

- return predictability becomes stronger exactly when the economy is weak.

Interpretation. This is an endogenous recession-amplification mechanism. The representative-agent model has no corresponding redistribution channel.

4.5 Why exogenous time-varying persistence is not enough

A representative-agent model with exogenous persistence shocks can match some movement in the persistence parameter, but it cannot replicate the same strong state dependence. In the heterogeneous-agent model, the same low wealth share for agent B implies very different future dynamics depending on whether x_t is high or low, because the demand for insurance depends on the current state.

Interpretation. The key object is not just “how persistent is growth risk today?” It is “who is wealthy today, and how does the current state change the value of insurance?” That interaction is inherently heterogeneous-agent.

4.6 Limits and scope

- The model is complete markets, so it isolates the pure pricing effects of heterogeneous beliefs without financial frictions.
- The paper is quantitative rather than econometric: it does not estimate the full heterogeneous-agent model from data.
- The mechanism relies on recursive preferences with a preference for early resolution of risk; without that feature the survival and pricing results weaken sharply.

Interpretation. The paper is best read as showing that investor heterogeneity is a missing state variable in modern long-run risk models, not as the final word on asset pricing.

5 Tables and figures

5.1 Table 1 and Figure 1: why disagreement about ρ_x is plausible

Read:

- Table 1 simulates how accurately an investor could estimate ρ_x from finite samples.
- Even after 60 years of monthly data, the 5% quantile of the estimate is still far below the true $\rho_x = 0.99$, especially when x_t is unobserved and must be filtered from consumption growth.
- Figure 1 uses Survey of Professional Forecasters data and shows a wide cross-sectional distribution of AR(1) persistence coefficients for consumption growth forecasts.
- The F-test for equal persistence beliefs is strongly rejected.

Economic message. The paper’s heterogeneity assumption is empirically grounded: long-run persistence is both hard to estimate and visibly disputed in professional forecasts.

Table 1

Parameter estimates from simulated data.

		s_t observable		s_t unobservable all parameters
		with constant	w/o constant	
60 years	$\hat{\rho}_t$	0.9836	0.9872	0.9613
	$\hat{\rho}_{t,0.95}$	0.9699	0.9750	0.9116
	$\hat{\rho}_{t,0.05}$	0.9583	0.9641	0.5120
100 years	$\hat{\rho}_t$	0.9865	0.9886	0.9813
	$\hat{\rho}_{t,0.95}$	0.9764	0.9803	0.9566
	$\hat{\rho}_{t,0.05}$	0.9713	0.9756	0.9154
500 years	$\hat{\rho}_t$	0.9893	0.9897	0.9887
	$\hat{\rho}_{t,0.95}$	0.9862	0.9865	0.9828
	$\hat{\rho}_{t,0.05}$	0.9844	0.9847	0.9783

The table shows the mean point estimates of ρ_t , as well as the 5% and 95% quantiles after 60, 100, and 500 years obtained from simulating 1000 monthly time series of data. In the first approach, Eq. (19) is used for s_t , assuming the process is directly observable, and least squares is used to estimate the model parameters; we distinguish two cases of estimating the AR(1) model, the case with and the case without a constant. The second approach assumes that s_t is unobservable, and the full state-space model (20) is estimated using the Kalman filter. For the data-generating process, we use $\mu_A = 0$, $\rho_A = 0.99$, $\sigma_A = 0.0003432$, $\mu_B = 0.0015$, and $\sigma = 0.0078$.

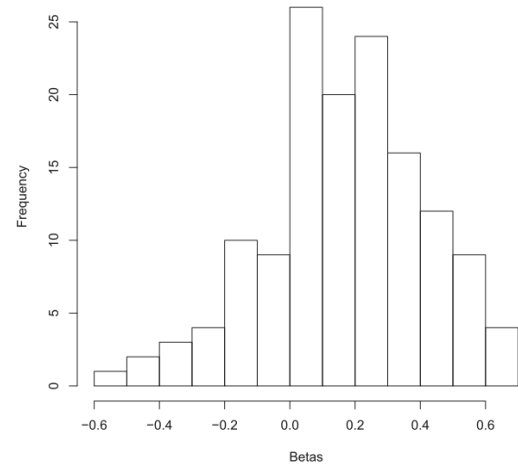


Fig. 1. Individual forecaster AR(1) coefficients. The figure shows a histogram of the AR(1) coefficients ($\beta + \beta_i$) for real consumption growth forecasts (see model (21)) for a group of forecasters from the US Survey of Professional Forecasters. Only forecasters with predictions in at least eight surveys are included in the sample. Regressions have been performed with quarterly fixed effects. For the presentation of the histogram, the two highest and the two lowest coefficients have been removed.

5.2 Figure 2 and Figure 3: the mean–variance trade-off and the two motives for trade

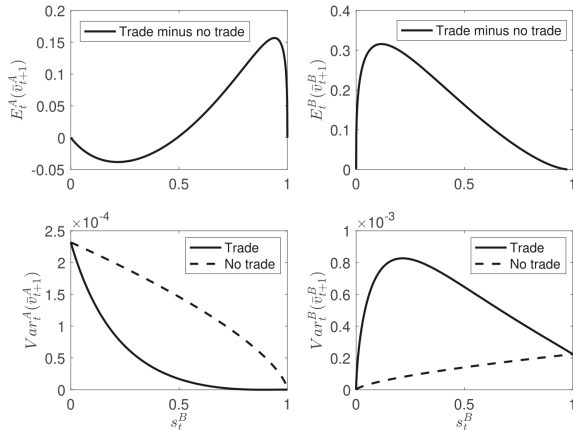


Fig. 2. Conditional mean and variance of continuation utilities. The figure plots the conditional mean and the variance of the continuation utilities as a function of the consumption share of investor B for the discrete-state economy. The top graphs show the difference in expected utility between an economy in which the agents are allowed to trade and an economy with no trade. The lower panel shows the conditional variance for both, the trade and no-trade case. The left graphs show the results for investor A, who believes in higher persistence, and the right graphs show the results for investor B, who believes in lower persistence. The results are shown for economy being in the bad state 1.

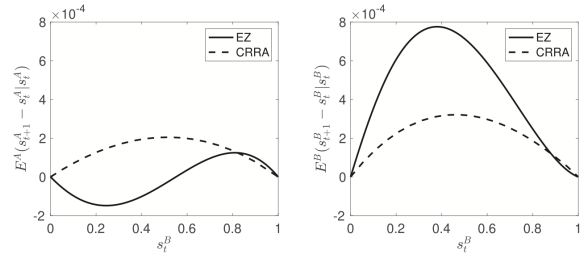


Fig. 3. Expected change in the consumption shares in the discrete-state model. The figure shows the expected changes in the consumption shares $s_{t+1}^h - s_t^h$, $h \in \{A, B\}$, as a function of s_t^B for the stylized economy with a two-state Markov chain for s_t . The left graph shows the expected change in the consumption share of agent A under the subjective beliefs of agent A. The right graph shows the expected change in the consumption share of agent B under the subjective beliefs of agent B. The solid lines show the expected changes for Epstein–Zin preferences, and the dashed lines show the corresponding results for CRRA preferences.

Read:

- Figure 2 compares trade and no-trade allocations in the discrete-state economy.
- Investor B , who believes in lower persistence, is willing to absorb more continuation-value variance.
- Figure 3 contrasts CRRA and Epstein–Zin dynamics for expected changes in consumption shares.
- Under CRRA, both investors make speculative bets; under Epstein–Zin, the investor who dislikes recession states more strongly is willing to transfer wealth in exchange for insurance.

Economic message. These figures are the cleanest visual demonstration that recursive preferences create a risk-sharing channel absent under CRRA. That channel is what allows wrong-belief investors to matter for prices.

5.3 Figures 4 and 5: state-dependent redistribution in consumption and wealth

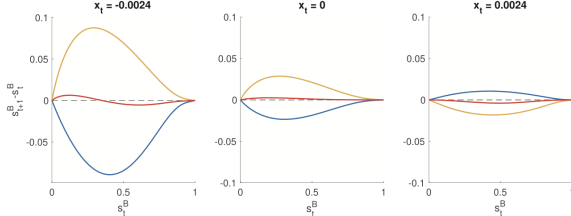


Fig. 4. Changes in the consumption shares for $\rho_A = \rho_A^1 = 0.99$ and $\rho_B^2 = 0.96$. The figure shows the change in the consumption share $s_{t+1}^B - s_t^B$ as a function of s_t^B . Results are shown for $x_t = 0$ and for ± 1 unconditional standard deviations of x_t around its unconditional mean of $x_t = 0$. The blue lines depict the case of a negative shock in x_{t+1} ($x_{t+1} - \rho_A x_t = -0.0005$), the yellow lines that of a positive shock in x_{t+1} ($x_{t+1} - \rho_A x_t = 0.0005$), and the red lines show the average over all shocks. Agent A has the correct beliefs with $\rho_A^1 = \rho_A = 0.99$, and agent B has the belief $\rho_B^2 = 0.96$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

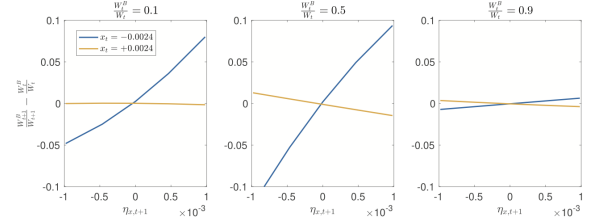


Fig. 5. Changes in the wealth shares for $\rho_A = \rho_A^1 = 0.99$ and $\rho_B^2 = 0.96$. The figure shows the change in the wealth share of investor B, $W_{t+1}^B - W_t^B$, as a function of the shock to x_t , x_{t+1} . The three graphs show the results for $W_t^B \in [0.1, 0.5, 0.9]$, respectively. The blue lines depict the case of a bad x -state ($x_t = -0.0024$) and the yellow lines that of a good x -state ($x_t = 0.0024$). Agent A has the correct beliefs with $\rho_A^1 = \rho_A = 0.99$, and agent B has the belief $\rho_B^2 = 0.96$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Read:

- Figure 4 shows how shocks move agent B 's consumption share as a function of the current state x_t and the current share s_t^B .
- In bad times, the movements are much larger because both the speculation motive and the risk-sharing motive become strong.
- Figure 5 translates this into changes in wealth shares. Investors with lower perceived persistence load more heavily on states in which mean reversion is expected.

Economic message. The redistribution channel is strongly nonlinear and state dependent. It is especially powerful in recessions, which is why the model produces countercyclical expected risk premia.

5.4 Figure 6 and Table 2: prices, moments, and endogenous excess volatility

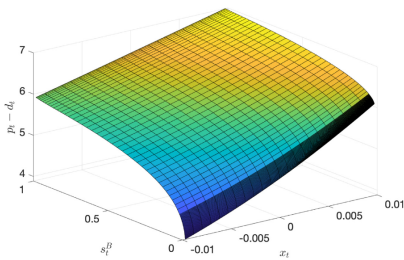


Fig. 6. Log price-dividend ratio for $\rho_A = \rho_A^1 = 0.99$ and $\rho_B^2 = 0.96$. The figure plots the graph of the log price-dividend ratio $p_t - d_t$ as a function of x_t and s_t^B . Agent A has the correct beliefs with $\rho_A^1 = \rho_A = 0.99$, and agent B has the belief $\rho_B^2 = 0.96$.

Read:

- Figure 6 plots the log price-dividend ratio as a function of x_t and the share s_t^B .
- Prices are higher when investor B has a larger share because that investor requires a smaller premium for holding risky dividends.

Table 2
Annualized asset-pricing moments.

	$E(p - d_t)$	$\sigma(p - d_t)$	$AC1(p - d_t)$	$E(R^e - R^f)$	$E(R^f)$	$\sigma(R^e)$	$\sigma(R^f)$
$s_t^B = 0$	2.36	0.28	0.79	12.52	1.72	18.66	1.65
Two-Ag	3.52	0.38	0.80	9.42	2.29	21.79	1.92
$s_t^B = 1$	3.97	0.11	0.38	1.78	2.93	14.08	0.88
Data	3.36	0.45	0.87	7.09	0.57	20.28	2.86

The table shows selected moments from 1000 samples—each containing 77 years of simulated data—starting with an initial share of $s_t^B = 0.5$. It reports the mean, the standard deviation, and the first-order autocorrelation of the annualized log price-dividend ratio as well as the mean and the standard deviation of both the annualized equity premium and the risk-free return. Agent A has the correct beliefs with $\rho_A^1 = \rho_A = 0.99$, agent B has the belief $\rho_B^2 = 0.96$. All returns are shown in percentages, so a value of 1.5 is a 15% annualized figure. The estimates from the data are taken from Biais et al. (2012).

- Table 2 shows that the heterogeneous-agent model produces a much more volatile price-dividend ratio and stock return than either representative-agent benchmark.
- The model's $\sigma(p_t - d_t)$ is about 0.38, much closer to the data value 0.45.

Economic message. The wealth distribution is an endogenous amplifier of price volatility. The model creates excess volatility without relying on exogenous stochastic volatility.

5.5 Figure 7 and Table 3: large countercyclical variation in expected risk premia

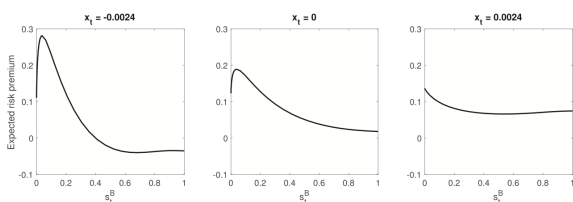


Fig. 7. Time variation in expected risk premia. The figure shows the annualized expected risk premium as a function of s_t^B . Results are shown for $x_t = 0$ and for ± 1 unconditional standard deviations of x_t around its unconditional mean of $x_t = 0$. Agent A has the correct beliefs with $\rho_x^A = \rho_x = 0.99$, and agent B has the belief $\rho_x^B = 0.96$.

Table 3
Expected risk premia.

	Data	Het. agents	BY (2004)	BKY (2012)
Mean	5.00	5.42	5.59	6.38
Std. dev.	4.60	5.73	1.13	1.61

The table shows the annualized mean and volatility of the expected risk premium. The first column shows the empirical values reported in Table 1 of Martin (2017). The second column shows the results for the heterogeneous-agent setup with $\rho_x^A = \rho_x = 0.99$ and $\rho_x^B = 0.96$ for the correct beliefs. The data are obtained from 1000 samples—each containing 77 years of simulated data—starting with an initial share of $s_0^B = 0.5$. Columns 3 and 4 show the results for the models of Bansal and Yaron (2004, Case II) and Bansal et al. (2012), respectively, using the same size for the simulated data set. All returns are shown in percentages, so a value of 1.5 is a 1.5% annualized figure.

Read:

- Figure 7 shows that expected risk premia depend strongly on the share s_t^B , especially when x_t is negative.
- Risk premia are highest when the high-persistence agent dominates and the economy is in a bad state.
- Table 3 shows that the model's standard deviation of expected risk premia is 5.73%, close to the data and far above the standard representative-agent long-run risk benchmark.

Economic message. The model explains Martin's large time variation in expected risk premia through endogenous redistribution rather than through exogenous volatility shocks.

5.6 Tables 4 and 5: predictability of returns and stronger recession predictability

Table 4
Predictability of excess returns, consumption, and dividends.

	R^2			β		
	1Y	3Y	5Y	1Y	3Y	5Y
$\sum_{i=0}^h (r_{t+i,h} - r_{f,t+i}) = \alpha + \beta(p_t - d_t) + \epsilon_{t+h}$						
$s_t^B = 0$	0.0070	0.0189	0.0277	-0.0134	-0.0406	-0.1116
Two-Ag.	0.0129	0.0350	0.0540	-0.0563	-0.1676	-0.2639
$s_t^B = 1$	0.0067	0.0113	0.0150	-0.0226	-0.0824	-0.1215
Data	0.04	0.19	0.31	-0.09	-0.27	-0.43
$\sum_{i=0}^h (\Delta d_{t+i}) = \alpha + \beta(p_t - d_t) + \epsilon_{t+h}$						
$s_t^B = 0$	0.5121	0.5738	0.5075	0.0829	0.2187	0.3124
Two-Ag.	0.3043	0.3346	0.2825	0.0495	0.1283	0.1805
$s_t^B = 1$	0.1522	0.1438	0.1012	0.0930	0.1941	0.2165
Data	0.060	0.01	0.000	0.01	0.01	0.00
$\sum_{i=0}^h (\Delta c_{t+i}) = \alpha + \beta(p_t - d_t) + \epsilon_{t+h}$						
$s_t^B = 0$	0.4881	0.4903	0.4393	0.3044	0.7074	0.9705
Two-Ag.	0.3088	0.2868	0.2486	0.1882	0.4233	0.5910
$s_t^B = 1$	0.4283	0.2051	0.1305	0.6557	0.9187	0.9977
Data	0.09	0.06	0.04	0.07	0.11	0.09

The table reports R^2 statistics and regression coefficients from regressing cumulative log excess returns, consumption growth, and dividend growth on the lagged log price-dividend ratio from 1000 samples—each containing 77 years of simulated data—starting with an initial share of $s_0^B = 0.5$. Statistics are shown for the annualized time series with one-, three-, and five-year horizons. The estimates from the data are taken from Bansal et al. (2012).

Table 5
Predictability in recessions and expansions.

	R^2			β_1			β_2		
	1Y	3Y	5Y	1Y	3Y	5Y	1Y	3Y	5Y
Recession dummy									
$s_t^B = 0$	0.0180	0.0393	0.0538	-0.0085	-0.0377	-0.0876	-0.0035	-0.0122	-0.0058
Two-Ag.	0.0475	0.0880	0.1182	-0.1016	-0.2465	-0.3884	-0.0259	-0.0396	-0.0426
$s_t^B = 1$	0.0190	0.0287	0.0357	-0.0217	-0.0761	-0.1121	-0.0001	-0.0003	-0.0005
Expansion dummy									
$s_t^B = 0$	0.0188	0.0395	0.0538	-0.0075	-0.0224	-0.0666	-0.0017	-0.0066	-0.0104
Two-Ag.	0.0245	0.0597	0.0941	-0.0621	-0.1896	-0.2898	0.0084	0.0210	0.0336
$s_t^B = 1$	0.0199	0.0303	0.0378	-0.0190	-0.0773	-0.1233	-0.0003	-0.0004	-0.0021

The table reports R^2 statistics and regression coefficients from regressing cumulative log excess returns on the lagged log price-dividend ratio including a recession (expansion) slope dummy. $\sum_{i=0}^h (r_{t+i,h} - r_{f,t+i}) = \alpha + \beta_1(p_t - d_t) + \beta_2 d_{\text{recession}}(p_t - d_t) + \epsilon_{t+h}$. Median values are shown for 1000 samples—each containing 77 years of simulated data—starting with an initial share of $s_0^B = 0.5$. Statistics are shown for the annualized time series with one-, three-, and five-year horizons.

Read:

- Table 4 shows that the heterogeneous-agent model increases return predictability while reducing the excessive predictability of consumption and dividend growth present in the representative-agent long-run risk model.
- Table 5 shows that return predictability is significantly stronger in recessions than in expansions.
- This occurs because recession shocks shift wealth toward the high-persistence agent, raising risk prices exactly when the economy is weak.

Economic message. The model delivers both the level and the cyclical variation of predictability. That is a major advantage over the standard long-run risk model.

5.7 Figure 8 and Table 6: why an exogenous time-varying-persistence model is not enough

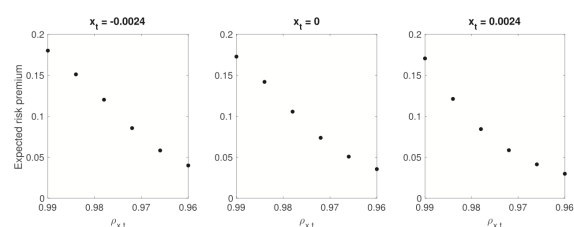


Fig. 8. Time variation in expected risk premium in the model with time-varying persistence and a negative correlation. The figure shows the annualized expected risk premium as a function of $\rho_{x,t}$ for the representative-agent model with time-varying persistence and a negative correlation between x_t and $\rho_{x,t}$. Results are shown for $x_t = -0.0024$, $x_t = 0$ and $x_t = 0.0024$.

Table 6
Expected risk premia in the model with time-varying persistence.

	Data	Het. agents	TV pers.
Mean	5.00	5.42	8.80
Std. dev.	4.60	5.73	2.49

The table shows the annualized mean and volatility of the expected risk premium. The first column shows the empirical values reported in Table 1 of Martin (2017). The second column shows the results for the heterogeneous-agent setup with $\rho_x^A = \rho_x = 0.99$ and $\rho_x^B = 0.96$ for the correct beliefs. Column 3 shows the results for the representative-agent model with time-varying persistence. All returns are shown in percentages, so a value of 1.5 is a 1.5% annualized figure.

Read:

- Figure 8 plots expected risk premia in a representative-agent model with exogenous time variation in persistence.
- Risk premia move with $\rho_{x,t}$, but they are much less sensitive to the current state x_t than in the heterogeneous-agent model.
- Table 6 confirms that the volatility of expected risk premia is much lower than in the heterogeneous-agent economy.

Economic message. Endogenous redistribution and exogenous persistence variation are not equivalent. The wealth-distribution channel adds an extra, state-dependent force that the representative-agent alternative misses.

5.8 Figure 11: sensitivity to risk aversion and belief gaps

Read:

- Figure 11 studies how the long-run distribution of wealth changes when the degree of risk aversion and the belief gap vary.
- Higher risk aversion strengthens the risk-sharing motive and can transfer wealth toward the lower-persistence investor even when that investor is wrong.

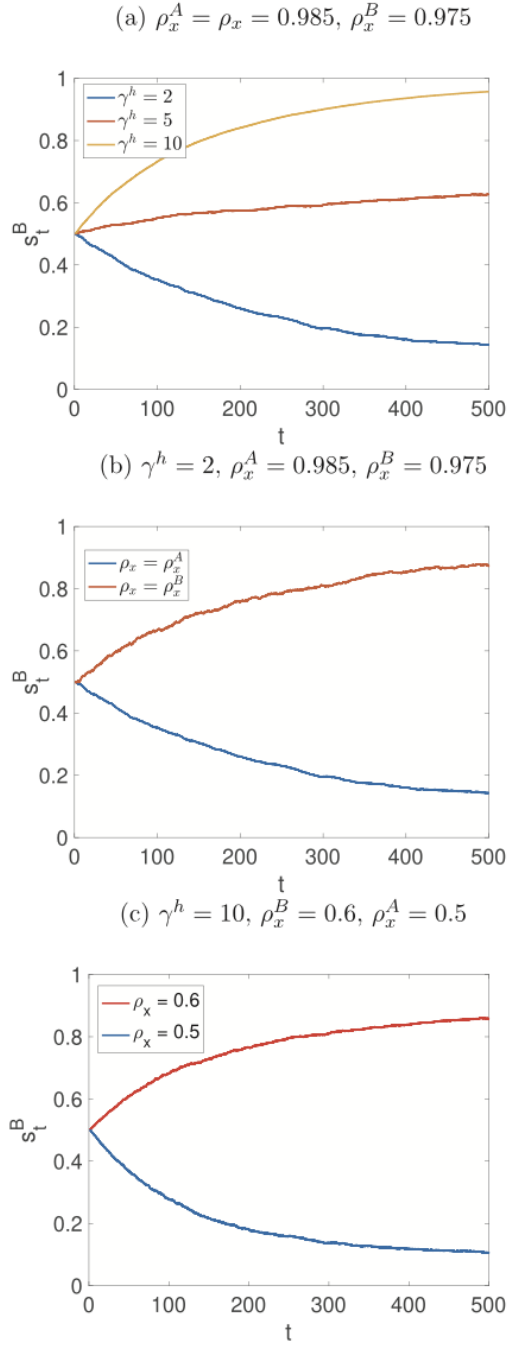


Fig. 11. The risk-sharing and speculation motives. The figure shows the median consumption share of agent B for 1000 samples, each consisting of 500 years of simulated data. Panel (a) shows the time series for different degrees of risk aversion $\gamma^h \in \{2, 5, 10\}$. Agent B believes that $\rho_x^B = 0.975$, and agent A has the correct beliefs with $\rho_x^A = \rho_x = 0.985$. Panel (b) shows the time series for $\gamma^h = 2$, $\rho_x^A = 0.985$, and $\rho_x^B = 0.975$ for the two cases in which either agent A (blue line) or agent B (red line) has the correct beliefs. Panel (c) shows the time series for $\gamma^h = 10$, $\rho_x^A = 0.6$, and $\rho_x^B = 0.5$ for the two cases where either agent A (blue line) or agent B (red line) has the correct beliefs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- Lower risk aversion or much lower persistence makes the speculation motive dominate, so the investor with correct beliefs tends to take over.

Economic message. The model’s asset-pricing implications come from a balance between speculation and insurance. That balance is strongest exactly in the parameter region relevant for long-run risk asset pricing.

6 Short summary

- The paper extends the long-run risk model to a complete-markets economy with heterogeneous agents and recursive preferences.
- The main new state variable is the endogenous wealth distribution, summarized by time-varying Negishi weights and consumption shares.
- Agents disagree about the persistence of long-run risk, ρ_x , a parameter that is both difficult to estimate and empirically disputed.
- Consumption-share dynamics can be decomposed into a speculation term and an additional Epstein–Zin risk-sharing term.
- Under CRRA, the classical market-selection logic dominates; under Epstein–Zin, incorrect-belief investors can survive and affect prices because they provide insurance against long-run utility risk.
- The baseline two-agent model, which differs from Bansal–Yaron only in beliefs about ρ_x , produces a realistic equity premium, much more volatile price-dividend ratios, and large countercyclical variation in expected risk premia.
- The model also increases return predictability, lowers the excessive predictability of cash flows, and generates stronger return predictability in recessions.
- A representative-agent model with exogenous time-varying persistence cannot reproduce the same effects because it lacks the endogenous redistribution channel.
- The paper’s main lesson is that investor heterogeneity is not a minor detail in recursive asset-pricing models; it is a central state variable for prices.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that belief heterogeneity can have first-order asset-pricing effects in long-run risk models with recursive preferences. Once investors disagree about the persistence of long-run growth risk, the equilibrium wealth distribution becomes a key state variable, and that state variable substantially improves the model’s ability to match financial market facts.

7.2 Economic intuition

Recursive utility makes investors care about the timing and persistence of bad states, not only about one-period consumption. Because of that, disagreement does more than create speculative trading: it creates insurance demand. The investor who is more sensitive to recession states is willing to transfer wealth to the investor who is more willing to bear that risk. Those transfers are persistent, state dependent, and priced in equilibrium. Asset prices therefore move with who is wealthy, not just with aggregate growth.

7.3 Takeaway

The deepest lesson of the paper is that disagreement over a hard-to-estimate macro-finance parameter can be a parsimonious and powerful explanation for excess volatility, countercyclical risk premia, and recession-dependent predictability. In modern long-run risk asset pricing, heterogeneity is not an inconvenient complication; it is part of the mechanism.